

General Instructions

- Please enter your name, Roll Number, Subject Code, Subject name in the Front Page of Answer sheet.
- Maximum Marks 15

Section A: One mark Questions 5x1= 5 Marks

1. What the term *sequential consistency* denotes? [1]
2. Differentiate between the CUDA API Functions `cudaMemcpy`, `cudaMemcpyAsync`, and `cudaMemcpyToSymbol`. [1]
3. What tool can be used to debug a CUDA program for runtime errors inside a CUDA kernel? How you use it? [1]
4. How many configuration parameters can be given inside the execution configuration syntax `<<<<>>>` of a CUDA kernel launch? What they denote? [1]
5. How to check whether a *device* (GPU) operation (like kernel call or data transfer between the *host*/CPU and the *device*) is successful or not in a CUDA program? [1]

Section B: Descriptive Questions 3x4= 12 Marks

6. Write a CUDA kernel that implements barrier for a CUDA kernel using `__syncthreads()` and `atomicCAS` API functions. Explain your code. [4]
7. Given below is pseudocode for the matrix multiplication. How the below code can be modified for better throughput?. Justify your answer with modified pseudocode(s) and its explanation. [4]

```
for (i=0;i<n;i++){
    for(j=0;j<n;j++){
        for(k=0;k<n;k++){
            C[i,j]=C[i,j]+A[i,k]*B[k,j];
        }
    }
}
```

8. (a) Write a pseudocode for the parallel Breadth first Search (BFS) computation with no atomic operations. Explain your pseudocode.

OR

- (b) Write pseudocode for work efficient parallel *exclusive-scan* operation. Explain your pseudocode. How shared memory of a GPU can be used for the operation.

[4]